

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA0301	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	B . Tech IT/2024	Date:	24.07.2026

Code of Conduct

I Dharmika S(192421094) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Dharmika S

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

100%
e

soln

Total spectrum = 45 MHz = 45,000 kHz, Full duplex
channel needs 2 simplex channels = 2×25
= 50 kHz

$$\text{channels} = \frac{45,000}{50}$$
$$= 900$$

Ans: 900

2)

soln

channels per cell = $\frac{\text{total channels}}{N}$

$$= \frac{560}{7}$$

$$= 80$$

Ans: 80

3)

soln

total voice channels = $\frac{\text{carrier} \times \text{slot}}{\text{carrier}}$

$$= 100 \times 8$$

$$= 800$$

Ans: 800

4)

soln

$$\text{Power ratio} = 80/4 = 20,$$

$$\text{Ratio in dB} = 10 \times \log_{10}(20)$$

$$= 13.01 \text{ dB}$$

$$= 13 \text{ dB}$$

5) soln

$$C/I \text{ (dB)} = \text{signal (dB)} - \text{interference (dBm)}$$

$$= -60 - (-92) = 32 \text{ dB}$$

$$\boxed{= 32 \text{ dB}}$$

6) soln

$$\text{Total channels} = \frac{42,000 \text{ kHz}}{300 \text{ kHz}}$$

$$= 140$$

$$\text{channels per cell} = 140/7 = 20$$

$$\boxed{= 20}$$

7) soln

Received power $\propto 1/d^n$. doubling distance with n

$$n = 4 \text{ reduces power by } 2^4 = 16 \times$$

$$\text{New Power} = \frac{64}{16}$$

$$= 4 \text{ W}$$

$$\boxed{\therefore 4 \text{ W}}$$

8) soln

with $n=2$ (inverse square law) doubling radius requires power increased by $2^2 = 4 \times$ to maintain same edge SN

$$\boxed{\therefore 4 \times}$$

9) soln

$$\text{Total channels in cluster} = \text{cluster size} \times \text{channels cell}$$

$$= 4 \times 15$$

$$= 60$$

$$\boxed{\therefore 60}$$

soln

$$v = 100 \text{ km/h} = 27.8 \text{ m/s}$$

$$\text{doppler shift} = v \times f / c = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8}$$

$$= 83.3 \text{ Hz}$$

$$\approx 83.3 \text{ Hz}$$

soln

$$\text{slot durations} = \frac{\text{frame time}}{\text{slots}}$$

$$= \frac{5 \text{ ms}}{8}$$

$$= 0.625 \text{ ms}$$

soln

$$\text{carriers} = \text{total spectrum / spacing}$$

$$= 24,000 \text{ KHz} / 200 \text{ KHz}$$

$$= 120$$

$$\therefore 120$$

soln

soln

$$\text{Tx Power in dBm} = 10 \log_{10} (8000 \text{ mW})$$

$$= 39.03 \text{ dBm}$$

$$\text{Received power} = 39.03 - 33$$

$$= 6.03$$

$$\text{dBm} \approx 6 \text{ dB}$$

14)

Soln

using the log-distance path loss model reference to 1m (consistent with = 120 dB at 2km for a nearby frequency band) scaling the distance to 2.5 km given approximate free-space path loss of = 123 dB

$$\therefore 123 \text{ dB}$$

15)

Soln

$$\begin{aligned} \text{total capacity} &= \frac{\text{calls}}{\text{cell} \times \text{cells}} \\ &= 250 \times 6 = 1500 \end{aligned}$$

$$\therefore 1500$$

16)

Soln

$$\begin{aligned} \text{Active calls supported} &= \text{arrival rate} \times \text{avg duration} \\ &= 4000 \text{ calls/sec} \times 180 \text{ sec} \end{aligned}$$

$$= 720,000$$

17)

Soln

$$\begin{aligned} \text{total bandwidth} &= \text{channel} \times \text{bandwidth / channel} \\ &= 150 \times 25 \text{ kHz} \\ &= 3750 \text{ kHz} \\ &= 3.75 \text{ MHz} \end{aligned}$$

soln
number of guard bands between 30 bands = 29

$$\text{wasted bandwidth} = 29 \times 40 \text{ kHz}$$

$$= 1160 \text{ kHz}$$

$$= 1.16 \text{ MHz}$$

soln

$$\text{Speed} = 90 \text{ km/h} = 25 \text{ m/s}$$

$$\text{cell diameter} = 2 \times 1.5 \text{ km}$$

$$= 3000 \text{ m}$$

$$\text{Time per cell} = 3000 / 25$$

$$= 120 \text{ s}$$

$$\text{Handoffs / hour} = 3600 / 120 = 30$$

soln

$$\begin{aligned} \text{total users} &= \text{regional centers} \times \text{base station / center} \\ &\quad \times \text{users / base station} \end{aligned}$$

$$= 8 \times 40 \times 250$$

$$= 80,000$$

21)

soln

$$\text{Approximate cell area} = R^2$$

$$= (0.4 \text{ km})^2$$

$$= 0.16 \text{ km}^2$$

$$\text{Number of cells} = \frac{400}{0.16}$$

$$= 2500$$

$$\therefore 2500$$

22)

soln

$$\text{channels} = \text{total spectrum} / \text{channel width}$$

$$= 6000 \text{ KHz} / 30 \text{ KHz}$$

$$= 200$$

$$\therefore 200$$

23)

soln

$$\text{Total call minutes} = 1200 \times 4 \times 2.5$$

$$= 12000 \text{ min/hr}$$

$$\text{Traffic in Erlangs} = 12000 / 60$$

$$= 200$$

24)

soln

$$\text{users} = \text{total bandwidth} / \text{Bandwidth per user}$$

$$= 40000 \text{ KHz} / 400 \text{ KHz}$$

$$= 100$$

25)

soln

$$\text{total capacity} = \text{users/cell} \times \text{cells}$$

$$= 600 \times 1200$$

$$= 720000$$

$$\text{Ans ; } 720,000$$

paging operation = $\frac{\text{total cells}}{\text{cells operation}}$

$$= \frac{200}{4}$$

$$= 50$$

27) Soln

offered traffic = arrival rate $\times$ avg duration

$$= 40,000 \times 3$$

$$= 1,20,000 \text{ erlangs}$$

$$\text{excessor capacity} = 120000 - 100000 = 20,000$$

$$\text{fraction blocked} = \frac{20,000}{120,000} = 0.17$$

28) Soln

$$\text{Roaming users} = 15\% \times 600,000$$

$$= 90,000$$

29) Soln

utilization = peak traffic / processing rate

$$= 9000 / 12,000 = 0.75$$

$$= 75\%$$

30) Soln

total channels = bandwidth / channel width

$$= \frac{48,000}{200}$$

$$= 240$$

$$\text{channels per cell} = 240 / 4 = 60$$

31) Soln

$$\text{channels} = \frac{\text{Total spectrum}}{\text{channel width}}$$

$$= \frac{30,000}{200}$$

$$\boxed{= 150}$$

32) Soln

$$\text{Successful handoffs} = 99\% \times 2500$$

$$\boxed{= 2475}$$

33) Soln

$$Tx \text{ power in dBm} = 10 \log_{10} (10,000 \text{ mW})$$

$$\approx 40.79 \text{ dBm}$$

$$\text{Received power} = 40.79 - 45$$

$$\approx -4.21 \text{ dBm}$$

$$\boxed{\approx -4 \text{ dBm}}$$

34) Soln

$$\text{total capacity} = \text{users / cell} \times \text{cells}$$

$$= 900 \times 7$$

$$\boxed{= 6300}$$

soln

$$\text{slot duration} = \frac{\text{frame time}}{\text{slots}}$$

$$= 4.8 \text{ ms} / 8$$

$$= 0.6 \text{ ms}$$

36) soln

$$v = 150 \text{ km/h} = 41.67 \text{ m/s}$$

$$\text{Doppler shift} = v \times f / c$$

$$= \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8}$$

$$= 250 \text{ Hz}$$

37) soln

$$\text{offered traffic} = 70,000 \times 4$$

$$= 280,000 \text{ Erlangs}$$

$$\text{Excess} = 250,000 - 220,000$$

$$= 30,000$$

$$\text{fraction Blocked} = \frac{30,000}{280,000}$$

$$= 0.11$$

38) soln

$$\begin{aligned}\text{Total using cluster} &= \text{user / cell} \times \text{cell in cluster} \\ &= 600 \times 9 \\ &= 5400\end{aligned}$$

39) soln

$$\begin{aligned}\text{calls dropped / sec} &= 0.4\% \times 60,000 \\ &= 240\end{aligned}$$

40) soln

$$\begin{aligned}\text{path loss (dB)} &= \text{Tx power (dBm)} - \text{Rx power (dBm)} \\ &= 43 - (-87) \\ &= 130 \text{ dB}\end{aligned}$$

41) soln

$$\begin{aligned}\text{total coverage} &= \text{coverage / station} \times \text{no. of station} \\ &= 2.5 \times 150 = 375 \text{ km}^2 \\ &\Rightarrow 375 \text{ km}^2\end{aligned}$$

42) soln

$$\begin{aligned}\text{Total channels} &= 48,000 \\ &\quad \underline{200} \\ &= 240\end{aligned}$$

$$\begin{aligned}\text{channels per cell} &= \frac{240}{6} \\ &= 40\end{aligned}$$

soln

$$\text{paging cycles} = \frac{\text{total users}}{\text{user per cycle}}$$

$$= \frac{1507000}{6000}$$

$$= 25$$

44) soln

$$\text{Handoffs per call} = \frac{\text{call duration}}{\text{handoff interval}}$$

$$= \frac{9}{3}$$

$$= 3$$

45) soln

$$\text{SNR (dB)} = \text{signal (dBm)} - \text{noise (dBm)}$$

$$= -85 - (-115)$$

$$= 30 \text{ dB}$$

46) soln

$$\text{Min bandwidth} = \text{users} \times \text{rate/user}$$

$$= 250 \times 150 \text{ kbps}$$

$$= 37,500 \text{ kbps}$$

$$= 37.5 \text{ Mbps}$$

48) soln

$$\text{Total call} = \frac{\text{user} \times \text{calls}}{\text{hr} \times \text{avg duration}}$$

$$= 1500 \times 4 \times 2.5$$

$$= 15000 \text{ min}$$

$$\text{Traffic in Erlangs} = 15000 / 60 = 250$$

48) soln

$$\text{Total bandwidth} = \frac{\text{channel} \times \text{bandwidth}}{\text{channel}}$$

$$= 250 \times 12 \text{ KHz}$$

$$= 3000 \text{ KHz}$$

$$= 3 \text{ MHz}$$

49) soln

$$\text{Total Throughput} = \frac{\text{Connections} \times \text{rate}}{\text{Connection}}$$

$$= 1500 \times 600 \text{ Kbps}$$

$$= 900,000 \text{ Kbps}$$

$$= 900 \text{ Mbps}$$

50) soln

$$\text{Area per tower} = \pi R^2 = \pi \times 4^2 = 50.27 \text{ km}^2$$

$$\text{Towers needed} = 350 / 50.27$$

$$= 6.96 \approx 7$$